IoT and Data Storage System Vulnerabilities

CompTIA PenTest+ (PT0-003)
Domains:

- 3.0: Vulnerability Discovery and Analysis
- 4.0: Attacks and Exploits

Objectives:

- 3.1: Given a scenario, conduct vulnerability discovery using various techniques.
- 4.1: Given a scenario, analyze output to prioritize and prepare attacks.

Video Overview

This video will explore the vulnerabilities of IoT devices and data storage systems, demonstrating how understanding these weaknesses is essential for protecting personal and professional networks. By examining real-world examples and potential attack vectors, you will gain insights into how these concepts apply to securing data in various environments, from smart homes to industrial control systems, and how this knowledge is vital for a career in cybersecurity.

Key Topics Covered

- IoT device proliferation and attack surface expansion
- Common IoT vulnerabilities: insecure defaults, weak credentials, clear-text communications
- Outdated firmware and lack of regular updates for IoT devices
- Specialized industrial system vulnerabilities (ICS, SCADA, IIoT) and attack types
- Tools for industrial system reconnaissance: Shodan.io, Censys, S7scan, PLCscan, SmartRF Packet Sniffer, ISF
- Data storage system vulnerabilities: misconfigurations, network exposure
- Improper permissions (e.g., "world read/write") on storage systems
- Lack of user input sanitization and verbose error messages in storage systems

Student Notes Section (Optional)